Quartic simplex interpolation: a tetrahedral Fejér theorem and a dimension-free fourth-power bound

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Abstract

For Bos's prescribed quartic interpolation nodes, we prove the Fejér inequality on a tetrahedron and the fourth-power inequality in every dimension, including the equality set. Finite exact-arithmetic witnesses and reconstructing verifiers are supplied. These resolve assertions 2 and 4 of AIM 641; the all-dimensional quartic Fekete assertion remains unresolved.

Catalogue target. AIM 641 at repository revision `c4036871b0bf`. The package preserves its full statement in `statement.md`; the review and target comparison are linked from the package README.

1 Quartic simplex interpolation: Problems 641(2) and 641(4)

We prove two assertions from Problem 641: the prescribed quartic nodes are Fejér points on a tetrahedron, and their cardinal polynomials satisfy the fourth-power inequality in every dimension. Both proofs reduce the required inequalities to explicit finite calculations in an algebraic number field. The accompanying standard-library Python verifiers reconstruct these calculations exactly. The known planar quartic Fejér theorem [1, Proposition 4.1], attributed there to [2], is the base case; we supply an independent elementary proof below, making the argument self-contained. The all-dimensional quartic Fekete assertion remains open in this report.

1.1 Nodes and homogeneous cardinal polynomials

Write $m = d + 1$ and use barycentric coordinates on

$$\Delta_{m-1} = \{x \in [0,\infty)^m : p_1(x) = 1\}, \qquad p_j(x) = \sum_{i=1}^{m} x_i^j.$$

Set

$$r = \sqrt{5}, \qquad q = \sqrt{3/7}, \qquad a = \frac{4+r}{11}, \qquad b = \frac{7-r}{22}.$$

*The solutions were found by AI. The submitted arguments and the repository audit are AI work; independent human review and formal verification are not claimed.

The node set $\mathcal{F}_{4,m-1}$ consists of all distinct coordinate permutations of the following vectors, padded with zero coordinates:

$$(1), \quad (\tfrac{1}{2},\tfrac{1}{2}), \quad (\tfrac{1+q}{2},\tfrac{1-q}{2}), \quad (a,b,b), \quad (\tfrac{1}{4},\tfrac{1}{4},\tfrac{1}{4},\tfrac{1}{4}). \tag{1.1}$$

An orbit is omitted when it has more than m nonzero coordinates. The number of nodes is

$$N_m = m + 3\binom{m}{2} + 3\binom{m}{3} + \binom{m}{4} = \binom{m+3}{4},$$

the dimension of polynomials of degree at most four on Δ_{m-1}.

We use the following degree-four homogeneous polynomials on $\mathbb{R}^m$. Their restrictions to $p_1 = 1$ are the cardinal polynomials. The vertex polynomials are

$$\begin{aligned} V_i = 2x_i\Big[&\frac{101+17r}{11}x_i^3 - (13+2r)x_i^2 p_1 + \frac{130+27r}{22}x_i p_1^2 - \frac{259+51r}{264}p_1^3 \\ &+ p_2\left(\frac{9+r}{8}p_1 - \frac{24+17r}{22}x_i\right) + p_3\left(\frac{45+3r}{44} - \frac{5}{3}\right)\Big]. \end{aligned} \tag{1.2}$$

For each unordered pair $\{i,j\}$, define

$$\begin{aligned} U_{ij} = {}& \frac{61+7r}{22}(x_i^2+x_j^2) + \frac{3+9r}{11}x_i x_j - \frac{41+13r}{22}(x_i+x_j)p_1 \\ &+ \frac{8+2r}{11}p_1^2 - \frac{7-r}{11}p_2, \end{aligned} \tag{1.3}$$

$$W_{ij} = \frac{7+r}{2}(x_j^2 - x_i^2) + \frac{5+r}{2}(x_i - x_j)p_1, \tag{1.4}$$

$$E_{ij}^{\pm} = \frac{49}{6}x_i x_j (U_{ij} \pm q W_{ij}), \tag{1.5}$$

$$\begin{aligned} M_{ij} = {}& \frac{4}{33}x_i x_j\big[(100-8r)(x_i^2+x_j^2) + (672-96r)x_i x_j \\ &+ (62r-302)(x_i+x_j)p_1 + (82-40r)p_2 + (76-14r)p_1^2\big]. \end{aligned} \tag{1.6}$$

Interchanging i,j interchanges the two labels $+$ and $-$. For each unordered triple there are three polynomials, one for each distinguished coordinate:

$$T_{i;jk} = \frac{73+r}{2}x_i x_j x_k\left[(1+r)x_i + \frac{3-r}{2}(x_j+x_k) - p_1\right]. \tag{1.7}$$

For each unordered quadruple there is one polynomial,

$$C_{ijkl} = 256 x_i x_j x_k x_l. \tag{1.8}$$

Let $\mathcal{L}_m$ denote this family of N_m polynomials.

Lemma 1.1 (Exact cardinal verification)**.** *The evaluation matrix of $\mathcal{L}_m$ at $\mathcal{F}_{4,m-1}$ is a permutation matrix. Consequently $\mathcal{F}_{4,m-1}$ is unisolvent, and the restrictions of $\mathcal{L}_m$ to Δ_{m-1} are its cardinal polynomials.*

Proof. For $m = 4$, direct substitution in (1.2)–(1.8) gives all $35^2 = 1225$ entries exactly. This calculation is reproduced by `verify_641_exact.py`, using rational arithmetic in $\mathbb{Q}(r,q)$. There are 35 ones, one in each row and column, and all other entries are zero.

This finite check proves the assertion for arbitrary m. Every node has at most four nonzero coordinates. Every polynomial involving a distinguished coordinate that is zero at the node vanishes there. The remaining polynomial formulas depend only on their distinguished coordinates and p_1, p_2, p_3, and so agree exactly with their four-coordinate versions after zero coordinates are added or deleted. Thus every possible evaluation is already covered by the $m = 4$ check. The node count is the dimension of the polynomial space, which completes the assertion. □

Remark 1.2 (A correction verified by interpolation). The displayed midpoint formula in the public 2022 preprint of [1] omits the term $(76-14\sqrt{5})p_1^2$ in (1.6); without it, the midpoint value is $(14\sqrt{5}-43)/33$ instead of 1. A paired-edge formula also repeats one coordinate index. We use the formulas above, whose values are checked independently at every node. No claim is made here about whether these typographical issues persist in the final journal version.

1.2 An independent proof of the known planar base case

The planar result is already known and is not counted as a new resolution. The following short proof removes any need to assume its original algebraic certificate.

Lemma 1.3 (Planar quartic Fejér inequality). *For every $x \in [0,\infty)^3$,*

$$F_3(x) := (x_1+x_2+x_3)^8 - \sum_{L\in\mathcal{L}_3} L(x)^2 \geq 0.$$

Proof. Let $e_1 = x_1+x_2+x_3$, $e_2 = x_1x_2+x_1x_3+x_2x_3$ and $e_3 = x_1x_2x_3$. Direct expansion of the cardinal formulas gives

$$\begin{aligned} F_3 = {} & 20e_1^6e_2 - 360e_1^4e_2^2 + 2100e_1^2e_2^3 - 3920e_2^4 \\ & + e_3\left[\frac{4623-57r}{11}e_1^5 + \frac{-125333+7219r}{33}e_1^3e_2 + \frac{29764-2772r}{3}e_1e_2^2\right] \\ & + e_3^2\left[\frac{-136228+14134r}{11}e_1^2 + \frac{77200-3296r}{3}e_2\right]. \end{aligned} \tag{1.9}$$

For nonnegative coordinates, $e_2 \leq e_1^2/3$. The coefficient of e_2 in the last bracket is positive, so that bracket is at most

$$\frac{-376852+90950r}{99}e_1^2 < 0 \qquad (e_1>0).$$

Thus F_3 is concave as a function of e_3 when e_1, e_2 are fixed.

For fixed feasible e_1, e_2, the set of nonnegative coordinate triples is compact. At either extreme value of their product e_3, at least one coordinate is zero or two coordinates are equal. To see this, suppose all coordinates are positive and distinct. The gradients of e_1, e_2 are then independent, so the Lagrange multiplier equations at a product extremum have the form $x_2x_3 = \lambda + \mu(x_2+x_3)$ and its cyclic analogues. Subtracting two equations forces the remaining coordinate to equal μ; distinctness would force all three coordinates to equal μ, a contradiction. Degeneracy of the constraint gradients already means all three coordinates are equal. Concavity therefore reduces the desired inequality to a zero coordinate and to two equal coordinates.

Both restrictions have explicit factorizations:

$$F_3(u,v,0) = 20uv(u-v)^2(u^2-5uv+v^2)^2, \tag{1.10}$$

$$F_3(u,v,v) = 40uv\left(u-\frac{3+r}{2}v\right)^2 Q(u,v), \tag{1.11}$$

where

$$Q(u,v) = (u^2+\kappa uv+v^2)^2 + v^2(\alpha u^2+\beta uv+\gamma v^2) \tag{1.12}$$

with

$$\kappa = \frac{-4397+383r}{880}, \qquad \alpha = \frac{23382299+5935673r}{1161600},$$
$$\beta = \frac{-229907+88485r}{1320}, \qquad \gamma = \frac{39579-17344r}{165}.$$

Here $\alpha, \gamma > 0$ and

$$4\alpha\gamma - \beta^2 = \frac{-529859170431 + 237067014259r}{11979000} > 0.$$

All displayed signs follow from the elementary bounds $559/250 < \sqrt{5} < 9/4$. Hence the quadratic form in (1.12) is positive definite, and both (1.10) and (1.11) are nonnegative for $u, v \geq 0$. Concavity proves the result; the origin is immediate.

Every identity and algebraic sign in this proof is checked directly from the frozen cardinal formulas by `verify_641_planar_base.py`, using only the same exact standard-library arithmetic. In particular, the factorization is verified as a polynomial identity, rather than inferred from sampled values or a numerical semidefinite calculation. □

1.3 The tetrahedral Fejér inequality

Theorem 1.4 (Problem 641(2))**.** *For every $x \in \Delta_3$,*

$$\sum_{L \in \mathcal{L}_4} L(x)^2 \leq 1. \tag{1.13}$$

Hence the prescribed 35 *quartic nodes are also Fekete points on a tetrahedron.*

Proof. Define the degree-eight homogeneous deficit

$$F(x) = p_1(x)^8 - \sum_{L \in \mathcal{L}_4} L(x)^2.$$

The family is invariant under coordinate permutations, so F is symmetric. The radical q cancels between the two edge terms, and all coefficients of F belong to $\mathbb{Q}(\sqrt{5})$. It suffices to prove $F \geq 0$ on the nonnegative orthant.

Order the coordinates and write

$$x_1 = u + v + w + t, \qquad x_2 = v + w + t, \qquad x_3 = w + t, \qquad x_4 = t, \qquad u, v, w, t \geq 0.$$

Put

$$G(u, v, w, t) = F(u + v + w + t, v + w + t, w + t, t), \qquad R = G(u, v, w, t) - G(u, v, w, 0).$$

The following is the finite algebraic certificate:

$$\mathcal{Q}(u, v, w, t) = (u + v + w + t)^{12} R(u, v, w, t) \quad \text{has only nonnegative coefficients.} \tag{1.14}$$

More precisely, R has 116 nonzero coefficients and $\mathcal{Q}$ has 1536. Every nonzero coefficient of $\mathcal{Q}$ is at least 60, and each has the form

$$\frac{A_\alpha + B_\alpha \sqrt{5}}{363}, \qquad A_\alpha, B_\alpha \in \mathbb{Z}. \tag{1.15}$$

The coefficient of $u^{19}t$ is exactly 60.

The attached file `fejer_641_certificate_independent.json` records these integer pairs. The verifier regenerates them from the cardinal formulas; it does not assume the correctness of the stored coefficient list. For an explicit description of the computation, expand R by ordinary polynomial convolution in $\mathbb{Q}(\sqrt{5})$, then use the recurrence

$$c_\alpha^{(j+1)} = \sum_{s=1}^{4} c_{\alpha - e_s}^{(j)}, \qquad 0 \leq j < 12, \tag{1.16}$$

with missing or negative-index entries zero and $c^{(0)}$ the coefficients of R. Every sign is checked exactly by integer comparisons: for the mixed-sign cases of $A + B\sqrt{5}$, compare A^2 and $5B^2$. A separate rational enclosure check uses

$$\frac{2236067977499789}{10^{15}} < \sqrt{5} < \frac{2236067977499790}{10^{15}},$$

with both endpoint inequalities verified by squaring integers. These finite operations prove (1.14).

It follows that $R \geq 0$, because $u + v + w + t > 0$ away from the origin and the assertion is trivial at the origin. When $t = 0$, all polynomials involving coordinate four vanish, and the remaining family is exactly the planar quartic cardinal family. Lemma 1.3 gives $G(u, v, w, 0) \geq 0$ directly in homogeneous form. Hence $G \geq 0$, and symmetry proves $F \geq 0$.

For the Fekete conclusion, evaluate the cardinal family at any competing 35 nodes and call the resulting matrix L. Each row has Euclidean norm at most one by (1.13). Hadamard's inequality gives $|\det \mathsf{L}| \leq 1$. The determinant of L is the ratio of the competing Vandermonde determinant to the one at the prescribed nodes, for any fixed polynomial basis. Thus the latter nodes maximize the absolute Vandermonde determinant. □

1.4 A stable symmetric deficit for fourth powers

Theorem 1.5 (Problem 641(4)). *For every $m \geq 2$ and every $x \in \Delta_{m-1}$,*

$$\sum_{L \in \mathcal{L}_m} |L(x)|^4 \leq 1. \tag{1.17}$$

In particular, every individual quartic cardinal polynomial is bounded in absolute value by one, in every dimension.

We first construct one symmetric polynomial which works for every coordinate count. For $x \in [0, \infty)^m$, put

$$H_m(x) = p_1(x)^{16} - \sum_{L \in \mathcal{L}_m} L(x)^4.$$

There is a polynomial $H \in \mathbb{Q}(\sqrt{5})[p_1, \ldots, p_{16}]$, independent of m, with weighted degree 16, whose specialization is H_m. Here the weight of p_j is j. It has 163 nonzero power-sum monomials. This polynomial can be reconstructed from the displayed cardinal formulas using only

$$\begin{aligned}
\sum_i x_i^a &= p_a, \\
\sum_{i \neq j} x_i^a x_j^b &= p_a p_b - p_{a+b}, \\
\sum_{\substack{i,j,k \\ \text{distinct}}} x_i^a x_j^b x_k^c &= p_a p_b p_c - p_{a+b} p_c - p_{a+c} p_b - p_{b+c} p_a + 2p_{a+b+c}, \\
\sum_{i<j<k<l} x_i^4 x_j^4 x_k^4 x_l^4 &= \frac{p_4^4 - 6p_4^2 p_8 + 3p_8^2 + 8p_4 p_{12} - 6p_{16}}{24}.
\end{aligned} \tag{1.18}$$

The exponents a, b, c in the distinguished variables are positive. For the paired-edge and midpoint contribution, apply the ordered-pair identity to $(E_{ij}^+)^4 + (E_{ij}^-)^4 + M_{ij}^4$ and divide by two. For the triple contribution, apply the ordered-triple identity to $T_{i;jk}^4$ and divide by two, since its two nondistinguished indices are interchangeable. The last line treats the quadruple contribution. As before, the radical q cancels.

Define formal derivatives in the polynomial ring by

$$A = \frac{\partial H}{\partial p_1}, \qquad B = \sum_{k=2}^{16} k p_{k-1} \frac{\partial H}{\partial p_k}. \tag{1.19}$$

The chain rule gives

$$D_m(x) := \sum_{i=1}^{m} \partial_{x_i} H_m(x) = mA(p(x)) + B(p(x)). \tag{1.20}$$

The next certificate is dimension independent.

Lemma 1.6 (Exact derivative certificate)**.** *For the power sums of any finite collection of nonnegative numbers,*

$$p_1^8 A \geq 0, \qquad p_1^8(5A + B) \geq 0. \tag{1.21}$$

Consequently $D_m(x) \geq 0$ for every $m \geq 5$ and $x \in [0, \infty)^m$.

Proof. For a partition $\lambda = (\lambda_1, \ldots, \lambda_\ell)$ into positive parts, let $m_\lambda^{(m)}$ be the monomial symmetric polynomial in m variables: each distinct monomial with nonzero exponent multiset λ occurs once. It is zero when $m < \ell$. The first exact expansion is

$$p_1^8 A = \sum_{\lambda \vdash 23} a_\lambda m_\lambda, \qquad a_\lambda \geq 0. \tag{1.22}$$

All 1255 partition types of 23 occur. For the second polynomial, write

$$p_1^8(5A + B) = \sum_{\lambda \vdash 23} b_\lambda m_\lambda.$$

Exactly eleven coefficients b_λ are negative. They are removed by the 28 same-length Muirhead transfers in Table 1.

The normalization is essential. If μ_j are the multiplicities of the distinct positive parts of λ, put

$$o_\lambda = \frac{\ell!}{\prod_j \mu_j!}, \qquad M_\lambda^{(m)} = \frac{m_\lambda^{(m)}}{o_\lambda}. \tag{1.23}$$

For each ℓ-element subset of the coordinates, the corresponding contribution to $M_\lambda^{(m)}$ is the average of all distinct permutations of the exponent vector λ. Therefore, if λ majorizes ν and they have the *same length*, Muirhead's inequality[3] applied on each such coordinate subset yields

$$M_\lambda^{(m)} \geq M_\nu^{(m)} \qquad \text{for every } m. \tag{1.24}$$

For m smaller than their common length both sides are zero. There is no dimension-dependent combinatorial factor in (1.24). This would generally be false if one transferred a normalized inequality between partitions of different lengths without correcting their multiplicities.

The exact identity is

$$p_1^8(5A + B) = \sum_{e=1}^{28} c_e (M_{\lambda_e} - M_{\nu_e}) + \sum_{\lambda \vdash 23} r_\lambda M_\lambda, \qquad c_e > 0, \quad r_\lambda \geq 0. \tag{1.25}$$

Every pair in the first sum has equal length and satisfies all the majorization prefix inequalities. There are nine length-three pairs, eighteen length-four pairs, and one length-five pair. The integers in Table 1 specify

$$c_e = \frac{A_e + B_e \sqrt{5}}{395307}.$$

The coefficients in (1.22), the coefficients b_λ, and the residual coefficients r_λ have the same common denominator and are all supplied in `fourth_641_muirhead_certificate.json`. Equivalently, the residuals are uniquely determined from the table by

$$r_\lambda = o_\lambda b_\lambda - \sum_{e:\lambda_e=\lambda} c_e + \sum_{e:\nu_e=\lambda} c_e. \tag{1.26}$$

There are 1227 nonzero residual coefficients.

For reproducibility, the verifier expands H using (1.18), differentiates to obtain A, B, and converts power-sum products to monomial coefficients by the following elementary recurrence. If C_μ is the coefficient of m_μ, the coefficient of m_ν in $p_k \sum C_\mu m_\mu$ is

$$\sum_{j:\nu_j \geq k} C_{\mathrm{sort}(\nu - ke_j)}. \tag{1.27}$$

Zero parts are deleted and equal positions are counted separately. Indeed, consider the coefficient of one fixed monomial: its factor from p_k can be supplied by any coordinate whose exponent is at least k. Repeated multiplication with $k = 1$ gives the factor p_1^8. The verifier checks every coefficient of (1.22), every majorization relation in the table, every positive payment, every residual sign in (1.26), and the complete identity (1.25). All signs are decided by exact rational and integer arithmetic in $\mathbb{Q}(\sqrt{5})$.

Nonnegative monomial coefficients prove the first inequality in (1.21). Muirhead and (1.25) prove the second. Finally,

$$p_1^8 D_m = p_1^8(5A + B) + (m-5)p_1^8 A \geq 0 \qquad (m \geq 5).$$

For nonzero $x \geq 0$, division by $p_1^8 > 0$ is legitimate; at the origin the result follows directly by homogeneity. □

Proof of Theorem 1.5. Theorem 1.4, homogeneity, and restriction to faces give the base cases $m \leq 4$:

$$\sum_{L \in \mathcal{L}_m} L(x)^4 \leq \left(\sum_{L \in \mathcal{L}_m} L(x)^2 \right)^2 \leq p_1(x)^{16}.$$

Suppose $m \geq 5$ and the result holds for $m - 1$. Given $x \geq 0$, set

$$a = \min_i x_i, \qquad y = x - a\mathbf{1}.$$

At least one coordinate of y is zero. Lemma 1.6 implies

$$H_m(x) - H_m(y) = \int_0^a D_m(y + t\mathbf{1})\, dt \geq 0.$$

Delete a zero coordinate to obtain $\widehat{y}$. The cardinal formulas are stable under this deletion: terms with that distinguished coordinate vanish and all other terms agree. Hence

$$H_m(x) \geq H_m(y) = H_{m-1}(\widehat{y}) \geq 0.$$

Induction proves the homogeneous inequality in all dimensions. Restricting to $p_1 = 1$ proves (1.17). □

Corollary 1.7 (The equality set)**.** *For the quartic cardinal family in every dimension, equality in* (1.17) *holds exactly at the prescribed interpolation nodes.*

Proof. The frozen certificate has $a_{(23)} = 40$ and $r_{(23)} = 160$. Since $m_{(23)} = M_{(23)} = p_{23}$, the nonnegative expansions (1.22) and (1.25) give

$$p_1^8 D_m \geq 40(m-1)p_{23} > 0 \qquad (m \geq 5,\ x \geq 0,\ x \neq 0).$$

If x has at least five positive coordinates, first delete its zero coordinates, obtaining a k-variable point with $k \geq 5$ and strictly positive minimum coordinate. The preceding minimum-coordinate integration is then strict, so $H_k(x) > 0$.

If the support has at most four coordinates, the square-sum Fejér bound applies. Equality in

$$\sum L(x)^4 \leq \Big(\sum L(x)^2\Big)^2 \leq 1$$

forces exactly one cardinal value to be 1 or -1 and all others to vanish. Partition of unity forces that value to be 1. Reproduction of each barycentric linear coordinate then identifies x with the corresponding interpolation node. Conversely, the cardinal equations give equality at every node. □

1.5 Exact witness and verification scope

The table below is the entire Muirhead transfer witness. Its role is to remove the eleven negative monomial coefficients of $p_1^8(5A+B)$, leaving the nonnegative residuals given by (1.26). Partitions are written as lists of positive parts, and every row has total degree 23.

Table 1: The 28 exact Muirhead payments, $c_e = (A_e + B_e\sqrt{5})/395307$.

e	λ_e	ν_e	A_e	B_e
1	$(13,7,3)$	$(13,6,4)$	1498725325337592	4787655964691688
2	$(13,7,3)$	$(13,5,5)$	8536773533222352	−3672905515511760
3	$(13,8,2)$	$(13,5,5)$	10632178332256140	86958143772588
4	$(14,7,2)$	$(14,5,4)$	5877037193901192	−79287322949208
5	$(14,6,3)$	$(14,5,4)$	1634552277469224	849845887111560
6	$(14,6,3)$	$(13,5,5)$	2236877237687496	−901354494250296
7	$(13,9,1)$	$(13,5,5)$	1971824645149920	256388529713184
8	$(15,6,2)$	$(15,4,4)$	445045370515296	−99159682090224
9	$(15,6,2)$	$(13,5,5)$	−22491224378121804	10692363542691444
10	$(8,7,5,3)$	$(8,6,5,4)$	42053528057751292992	323944859138228352
11	$(8,7,5,3)$	$(8,5,5,5)$	29821869380033387424	−349721950787748864
12	$(8,7,5,3)$	$(7,7,5,4)$	15227149544040964512	−1045096302260394240
13	$(9,6,5,3)$	$(7,7,5,4)$	34890784624009578288	1842836406593505648
14	$(9,6,5,3)$	$(7,6,6,4)$	48888060142970622096	−2773939678701409200
15	$(9,7,4,3)$	$(7,6,6,4)$	66937003491622708368	−1294371475424784336
16	$(8,7,6,2)$	$(7,6,6,4)$	−60291372213196792 80	3087929711581169136
17	$(8,7,6,2)$	$(7,6,5,5)$	64746162104032292976	−1885735473280680240
18	$(10,6,4,3)$	$(7,6,5,5)$	49084587467364645744	−195796276763905296
19	$(9,7,5,2)$	$(7,6,5,5)$	48480702963344097792	−661283746644118176
20	$(8,6,6,3)$	$(7,6,5,5)$	41670056935663240176	438034863058464816
21	$(7,7,6,3)$	$(7,6,5,5)$	34233944530210993200	1531159251798766608
22	$(10,6,5,2)$	$(7,6,5,5)$	4453774166762859552	27706801221905184
23	$(10,6,5,2)$	$(6,6,6,5)$	33665566828687711536	−1354359951228246096
24	$(8,8,4,3)$	$(6,6,6,5)$	36283080992417756064	−838102952190264960
25	$(9,6,4,4)$	$(6,6,6,5)$	31927055780967653856	−271911363432951264
26	$(10,5,5,3)$	$(6,6,6,5)$	28779142391558571360	78345804440140896
27	$(9,8,4,2)$	$(6,6,6,5)$	−163447391863139040	1812146122265513472
28	$(8,6,5,3,1)$	$(6,6,5,5,1)$	44030842344222195840	−7861535751971828640

The verification requires Python 3 and its standard library only:

```
python3 verify_641_exact.py
python3 verify_641_planar_base.py
python3 verify_641_fourth_exact.py
```

The first program checks all cardinal evaluations, symmetry and homogeneity of the degree-eight deficit, and all 1536 coefficient signs. The planar program verifies the elementary symmetric identity, the two factorizations, concavity, and positivity of the residual quadratic form. The fourth-power program reconstructs the stable polynomial H, checks its 163 terms and the exact derivative identity, and verifies (1.22) and (1.25). It also compares the power-sum expression with direct cardinal evaluations at exact rational points with $2,3,4,5,6,9,17$ coordinates. These pointwise checks supplement the formal algebraic derivation; they are not used as a substitute for it.

The two results establish 641(2) and 641(4). The tetrahedral Fekete property follows as a corollary, but a fourth-power bound does not imply the Hadamard bound needed for the all-dimensional Fekete assertion 641(3). That assertion remains unresolved here. The cubic Fejér assertion 641(1) was already marked solved in the catalogue and is not counted as new work. The proofs above are computer-assisted exact proofs with explicit finite witnesses; they have not been checked by a formal proof assistant or external human peer review.

References

[1] L. Bos, *On Fekete points for a real simplex*, Indag. Math. **34** (2023), 274–293. Primary preprint: https://arxiv.org/abs/2205.06498; https://doi.org/10.1016/j.indag.2022.11.003.

[2] L. Bos, *Bounding the Lebesgue function for Lagrange interpolation in a simplex*, J. Approx. Theory **38** (1983), 43–59. https://doi.org/10.1016/0021-9045(83)90140-5. The planar Fejér result is restated in Proposition 4.1 of[1]. The 1983 full text was not independently retrieved; this report also supplies its own elementary proof of the needed planar case.

[3] R. F. Muirhead, *Some Methods applicable to Identities and Inequalities of Symmetric Algebraic Functions of n Letters*, Proc. Edinburgh Math. Soc. **21** (1902), 144–162. https://doi.org/10.1017/S001309150003460X.